

Introduction to Web Programming Course Project

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AI Declaration

For this project, I used Copilot to work on some code parts that I do not know the correct syntax, and to implement some ideas while I worked on the project (for example, interpolate temperature for missing data points as OpenWeather API only supports temperature in 3 hours interval for free account, and I wanted to fill all the data in the chart every hour).

Feature	Max points	Point I would like to get
Well written PDF report	3	3
Application is responsive and can be used on both desktop and mobile environment	4	4
Application works on Firefox, Safari, Edge and Chrome	3	3
The application has clear directory structure and everything is organized well	2	2
User can search for locations	1	1
User can use his/her location GPS-coordinates (Geolocation API)	2	2
At least two data/forecast providers are used (this means completely different data sources like	3	3

x.com and y.com, not just different API endpoints on same service)		
At least three data/forecast providers are used	2	
User sees the current weather at a specific location	1	1
User sees the forecast for the next 24 hour, hourly based	3	3
User sees the forecast for the next 7 days	3	3
All the weather forecast elements uses icons (and numbers) for e.g. sunny and cloudy weathers	3	3
The look and feel of the application reflects the current weather (e.g. it is blueish, when it is cold; reddish, when it is hot;, dark, when it is night...)	2	2
User sees simultaneously two forecast in a graph, e.g. there is temperature forecast for the next 24 hours and there are two lines telling how the data sources are providing (a bit) different data.	3	3* - please read the explanation for better understanding.
User has the option to tag some locations as her favorites and thus access them from the favorites menu	2	2
User has an option to switch between Celsius and Fahrenheit degrees and kelvins	2	2

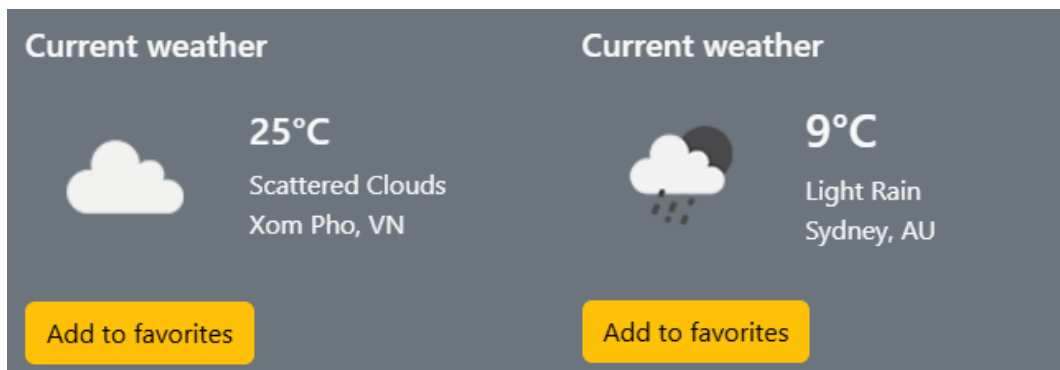
The tools used for the project:

- Visual Studio Code for coding
- Bootstrap for responsive and structured layout

- Frappe for visualization graph
- OpenWeather and WeatherAPI to provide data source for the application

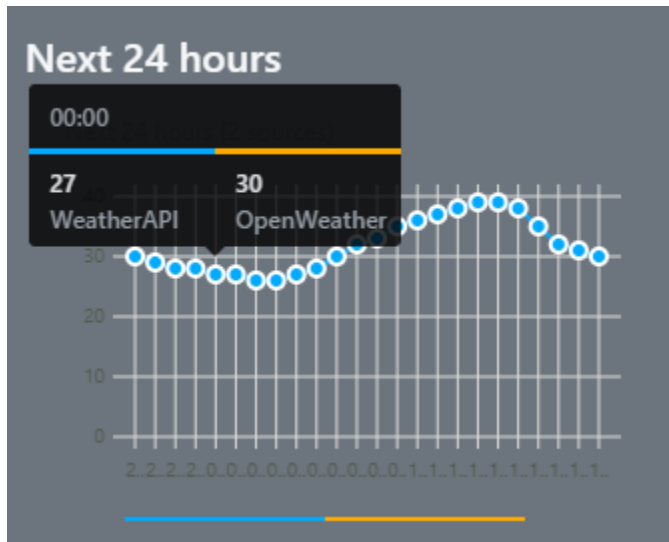
What was done and explanation for the application:

- The user can input the name of a city to see the current weather and forecast for next 24 hours/7 days. Input in the search field and click “Search”.
- User can also use his/her location to search by clicking “Use my location”.
- The result will display the current weather of the searched city, with icons for sunny/cloudy/rainy weather.

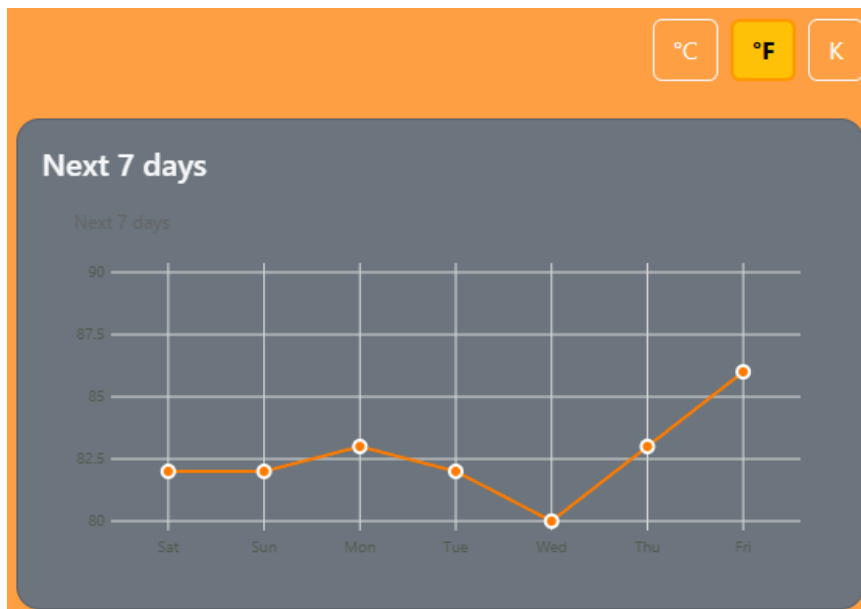


- Two data provider API were included in the project (OpenWeather as suggested from the course project description, but then I cannot get the hourly data for the graph, that's why I used WeatherAPI).
- 2 graphs show forecasts by hour in the next 24 hours and forecast by day for the next week.
- The background of the application changes according to the current temperature, as blue if it is cold and yellow/orange/red if it gets hotter. The exact color value can be found in line 266-274 in main.js.

- In the 24 hours forecast, there are two forecasts by two providers can be seen by hovering the mouse/pointer at the dots. As the data by OpenWeather only provide temperature by interval of 3 hours for free account only, the line shows the value from WeatherAPI only. For points with data from OpenWeather available, they can be seen next to WeatherAPI temperature. There are some dots with exactly same value but there are also dots where two sources provide different temperatures.



- User has option to switch between units. The default unit is Celsius degree, but if any unit is clicked, it will be toggled and change the graph accordingly, and the button will also have the visual changed to notify the user about the current unit.



- User can have their favorite cities included in the table next to search results for fast navigation. Favorite list can be updated/cleared with provided buttons.